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NANO-SHEET-BASED COMPLEMENTARY METAL-OXIDE-SEMICONDUCTOR DEVICES WITH ASYMMETRIC INNER SPACERS

Abstract

A semiconductor device is provided. The device includes a first pair and a second pair of source/drain features over a semiconductor substrate. The first pair of source/drain features are p-type doped. The second pair of source/drain features are n-type doped. A first stack of semiconductor layers connect the first pair of source/drain features along a first direction. A second stack of semiconductor layers connect the second pair of source/drain features along a second direction. A first gate is between vertically adjacent layers of the first stack of semiconductor layers. The first gate has a first portion that has a first dimension along the first direction. A second gate is between vertically adjacent layers of the second stack of semiconductor layers. The second gate has a second portion that has a second dimension along the second direction. The second dimension is larger than the first dimension.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation application of U.S. application Ser. No. 17/858,991, filed Jul. 6, 2022, which is a divisional application of U.S. application Ser. No. 16/875,726, filed May 15, 2020, which claims priority to U.S. Provisional Application No. 62/880,311 filed on Jul. 30, 2019, each of which is herein incorporated by reference in its entirety.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of IC processing and manufacturing, and for these advancements to be realized, similar developments in IC processing and manufacturing are needed.

[0003] For example, multi-gate devices have been introduced in an effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects (SCEs). One such multi-gate device is nano-sheet-based transistor, whose gate structure extends around its channel region providing access to the channel region on all sides. The nano-sheet-based transistors are compatible with conventional complementary metal-oxide-semiconductor (CMOS) processes, allowing them to be aggressively scaled down while maintaining gate control and mitigating SCEs. However, conventional methods for nano-sheet-based transistors may experience challenges in separately tuning current flow and capacitances of different device regions thereby falling short of achieving overall balanced and optimized performance. Therefore, although conventional nano-sheet-based transistors have been generally adequate for their intended purposes, they are not satisfactory in every respect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed

description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1A is a three-dimensional (3D) perspective view of an embodiment of a nano-sheet-based transistor that may be implemented as a PMOS and/or an NMOS in embodiments of nano-sheet-based CMOS devices of the present disclosure constructed according to some embodiments of the present disclosure.

[0006] FIG. 1B is a cross-sectional view of an embodiment of a nano-sheet-based transistor that may be implemented as a PMOS and/or an NMOS in embodiments of nano-sheet-based CMOS devices of the present disclosure along the line A-A' in FIG. 1A constructed according to some embodiments of the present disclosure.

[0007] FIGS. 2A-2D are flow charts of embodiments of methods for fabricating nano-sheet-based CMOS devices of the present disclosure according to some embodiments of the present disclosure.

[0008] FIGS. 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21A, 21B, 22, 23, 24, 25, 26, 27, 28A, and 28B are cross-sectional views of an embodiment of a PMOS and an embodiment of an NMOS in embodiments of nano-sheet-based CMOS devices of the present disclosure along the line A-A' in FIG. 1A constructed at various fabrication stages according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0009] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0010] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0011] Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term “about 5 nm” may encompass the dimension range from 4.5 nm to 5.5 nm.

[0012] The present disclosure is generally related to ICs and semiconductor devices and methods of forming the same. More particularly, the present disclosure is related to complementary metal-oxide-semiconductor (CMOS) device. CMOS is a type of semiconductor device that uses a combination of p-type transistors (PMOS) and n-type transistors (NMOS) to implement logic gates and other circuits. CMOS technology is used for constructing microprocessors, memory chips, microcontrollers, among others. As technologies progress towards smaller technology nodes (for example, 20 nm, 16 nm, 10 nm, 7 nm, 5 nm, and below), the optimization of CMOS performance increasingly hinges on reducing chip footprint and balancing performance characteristics of PMOS

and NMOS transistors. As a result, nano-sheet-based transistors have become a popular and promising architecture for building CMOS devices. The present disclosure provides methods of fabricating nano-sheet-based CMOS transistors. Particularly, the methods of the present disclosure decouple the fabrications of PMOS and NMOS transistors such that these different types of devices may be separately tuned for the optimal overall performance of the nano-sheet-based CMOS devices. For example, inner spacers of the PMOS and NMOS transistors of the present disclosure are made to have different physical dimensions, different surface profiles, and/or different material compositions such that the performance of the PMOS and NMOS devices can be separately tuned. [0013] A nano-sheet-based device includes any device that has a plurality of suspended gate channel layers stacked together to form the gate structure. Nano-sheet-based devices include gate-all-around (GAA) devices, multi-bridge-channel (MBC) devices, and other similar devices. Furthermore, the nano-sheet-based devices may include gate channel layers of any suitable shapes and/or configurations. For example, the gate channel layers may be in one of many different shapes, such as wire (or nanowire), sheet (or nanosheet), bar (or nano-bar), and/or other suitable shapes. In other words, the term nano-sheet-based devices broadly encompasses devices having gate channel layers in nanowire, nano-bars, and any suitable shapes. One of ordinary skill may recognize other examples of semiconductor devices that may benefit from aspects of the present disclosure. For example, other types of metal-oxide semiconductor field effect transistors (MOSFETs), such as planar MOSFETs, FinFETs, other multi-gate FETs may benefit from the present disclosure.

[0014] An exemplary nano-sheet-based transistor **100** is illustrated in FIGS. **1A** and **1B**. FIG. **1A** is a three-dimensional (3D) perspective view of an embodiment of a nano-sheet-based transistor that may be implemented as a PMOS and/or an NMOS transistor in embodiments of nano-sheet-based CMOS devices of the present disclosure. FIG. **1B** is a cross-sectional view of the nano-sheet-based transistor in FIG. **1A** along the line A-A'. As illustrated, the nano-sheet-based transistor **100** includes a semiconductor substrate **105**. Fin structures (or fins), such as fin structures **108**, are formed over the substrate **105**, each extending lengthwise horizontally in an X-direction and separated from each other horizontally in a Y-direction. It is understood that the X-direction and the Y-direction are horizontal directions that are perpendicular to each other, and that the Z-direction is a vertical direction that is orthogonal (or normal) to a horizontal XY plane defined by the X-direction and the Y-direction. The semiconductor substrate **105** may have its top surface parallel to the XY plane.

[0015] The fin structures **108** each have a source region, a drain region (collectively, source/drain regions **102**, or, source and drain regions **102**) disposed along the X-direction and spaced away from each other. Epitaxial source/drain features **210** are formed in the source/drain regions **102**. The fin structures **108** each further have a channel region **104** disposed between and connecting the source and drain regions **102**. A stack of suspended semiconductor layers **120** (also referred to as "channel layers **120**") are formed in the channel region **104** connecting the epitaxial source/drain features **210**; and the stack extends vertically (c.g. along the Z-direction) from the substrate **105**. The suspended semiconductor layers **120** may each be in one of many different shapes, such as wire (or nanowire), sheet (or nanosheet), bar (or nano-bar), and/or other suitable shapes, and may be spaced away from each other. In the depicted embodiments, there are three semiconductor layers **120** in the stack. However, there may be any appropriate number of layers in the stack, such as 2 to 10 layers. The semiconductor layers **120** may each engage with a single, contiguous gate structure **250**. Note that the gate structure **250** is illustrated as a transparent feature in FIG. **1A** in order to illustrate the features (such as the semiconductor layers **120**) that the gate structure **250** covers.

[0016] The nano-sheet-based transistor **100** further includes isolation features **150** within and/or over the substrate **105**, separating adjacent fin structures from each other. The isolation features **150** may be shallow trench isolation (STI) features. In some examples, the formation of the isolation features **150** includes etching trenches into the substrate **105** between the active regions (the

regions in which the fin structures are formed) and filling the trenches with one or more dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride, other suitable materials, or combinations thereof. Any appropriate methods, such as a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, a physical vapor deposition (PVD) process, a plasma-enhanced CVD (PECVD) process, a plasma-enhanced ALD (PEALD) process, and/or combinations thereof may be used for depositing the isolation features **150**. The isolation features **150** may have a multi-layer structure such as a thermal oxide liner layer over the substrate **105** and a filling layer (c.g., silicon nitride or silicon oxide) over the thermal oxide liner layer. Alternatively, the isolation features **150** may be formed using any other isolation technologies. As illustrated in FIG. **1A**, the fin structure **108** is located above the top surface of the isolation features **150**. The nano-sheet-based transistor **100** further includes gate spacer layers **200** on both sides of the gate structure **250**; optionally gate spacer layers **202** on two sides of the gate spacer layers **200**; optionally contact etch stop layers **220** over the epitaxial source/drain features **210**; interlayer dielectric (ILD) layer **230** over the epitaxial source/drain features **210** (and over the contact etch stop layers **220**, if present); and optionally source/drain spacers **207** over the isolation features **150**. [0017] FIGS. **2A-2D** are flow charts illustrating two embodiments of a method **10** for fabricating a nano-sheet-based CMOS device of the present disclosure according to some embodiments of the present disclosure. The first embodiment of method **10** (denoted as embodiment **10A**) and the second embodiment of method **10** (denoted as embodiment **10B**) share the same processing steps **12-30**, illustrated in FIGS. **2A** and **2B**. The embodiment **10A** continues from step (or block) **30** to **32A** and through **46A**, as illustrated in FIG. **2C**. The embodiment **10B** continues from step (or block) **30** to **32B** and through **44B**, as illustrated in FIG. **2D**. FIGS. **3-20**, **21A**, and **21B** are cross-sectional views of a nano-sheet-based CMOS device along the line A-A' in FIG. **1A** constructed at various fabrication stages according to the embodiment **10A** of method **10**; and FIGS. **22-27**, **28A**, and **28B** are cross-sectional views of a nano-sheet-based CMOS device along the line A-A' in FIG. **1A** constructed at various fabrication stages according to the embodiment **10B** of method **10**. [0018] Referring to block **12** of FIG. **2A** and FIG. **3**, the nano-sheet-based CMOS includes a substrate **105**. The substrate **105** contains a semiconductor material, such as bulk silicon (Si), germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), gallium arsenic (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb), or combinations thereof. The substrate **105** may also include a semiconductor-on-insulator substrate, such as Si-on-insulator (SOI), SiGe-on-insulator (SGOI), Ge-on-insulator (GOI) substrates. Some regions of the substrate **105** may be doped. For example, a region configured for a PMOS transistor of the nano-sheet-based CMOS (such as PMOS transistor **100A**) may be doped with n-type dopants, such as phosphorus (P) and/or arsenic (As); a region configured for an NMOS transistor of the nano-sheet-based CMOS (such as NMOS transistor **100B**) may be doped with p-type dopants, such as boron (B) and/or boron fluoride (BF₃). The doped regions may be formed directly on the substrate **105**, in a p-well structure, in an n-well structure, in a dual-well structure, or using a raised structure.

[0019] Referring to blocks **14** and **16** of FIG. **2A** and FIG. **3**, stacks of semiconductor layers are formed over the substrate **105** to later form the PMOS transistor **100A** and the NMOS transistor **100B**. The stacks of semiconductor layers include semiconductor layers **110** and semiconductor layers **120** alternating with each other. For example, a semiconductor layer **110** is formed over the substrate **105**; a semiconductor layer **120** is formed over the semiconductor layer **110**; and another semiconductor layer **110** is formed over the semiconductor layer **120**, so on and so forth. The material compositions of the semiconductor layers **110** and **120** are configured such that they have an etching selectivity in a subsequent etching process. For example, in some embodiments, the semiconductor layers **110** contain silicon germanium (SiGe), while the semiconductor layers **120** contain silicon (Si). In some other embodiments, the semiconductor layers **120** contain SiGe, while the semiconductor layers **110** contain Si. In the depicted embodiment, the semiconductor layers **110**

each have a substantially uniform thickness, referred to as the thickness **300**, while the semiconductor layers **120** each have a substantially uniform thickness, referred to as the thickness **310**. The thickness **310** may be the same as, or different from, the thickness **300**.

[0020] The stacks of semiconductor layers are patterned into a plurality of fin structures, for example, fin structures **108A** and **108B**, such that they each extend along the X-direction. The fin structures **108A** and **108B** may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins. The patterning may utilize multiple etching processes which may include a dry etching and/or wet etching. The fin structures **108A** and **108B** may have lateral widths (not shown) along the Y-direction that are the same between each other or different from each other.

[0021] Referring to block **18** of FIG. 2A and FIG. 4, dummy gate structures **240** are formed over a portion of each of the fin structures **108A** and **108B**. In some embodiments, the dummy gate structures are also formed over the isolation features **150** in between adjacent fin structures. The dummy gate structures **240** may be configured to extend lengthwise in parallel to each other, for example, each along the Y-direction. In some embodiments, the dummy gate structures each wrap around the top surface and side surfaces of each of the fin structures **108A** and **108B**. The dummy gate structures **240** may include polysilicon. In some embodiments, the dummy gate structures **240** also include one or more mask layers (not shown), which are used to pattern the dummy gate electrode layers. The dummy gate structures **240** may undergo a gate replacement process through subsequent processing to form metal gates, such as a high-k metal gate, as discussed in greater detail below. The dummy gate structures **240** may be formed by a procedure including deposition, lithography, patterning, and etching processes. The deposition processes may include CVD, ALD, PVD, other suitable methods, and/or combinations thereof.

[0022] Gate spacers are formed on the sidewalls of the dummy gate structures **240** and the top layer of the semiconductor layers **120**. Gate spacers may include a single layer or a multi-layer structure. For example, in the depicted embodiment, a gate spacer layer **200** is formed over the top surface of the device, and a gate spacer layer **202** is formed over the gate spacer layer **200**. The gate spacer layers **200** and **202** may each include silicon nitride (Si_3N_4), silicon oxide (SiO_2), silicon carbide (SiC), silicon oxycarbide (SiOC), silicon oxynitride (SiON), silicon oxycarbon nitride (SiOCN), carbon doped oxide, nitrogen doped oxide, porous oxide, or combinations thereof. In some embodiments, the gate spacer layers **200** and **202** may collectively have a thickness in the range of a few nanometers (nm). In some embodiments, the gate spacer layers **200** and/or **202** may be formed by depositing a spacer layer (containing the dielectric material) over the dummy gate structures **240**, followed by an anisotropic etching process to remove portions of the spacer layer from the top surfaces of the dummy gate structures **240**. After the etching process, portions of the spacer layer on the sidewall surfaces of the dummy gate structures **240** substantially remain and become the gate spacer layers **200** and/or **202**. In some embodiments, the anisotropic etching process is a dry (e.g. plasma) etching process. Additionally or alternatively, the formation of the gate spacer layers **200** and/or **202** may also involve chemical oxidation, thermal oxidation, ALD, CVD, and/or other suitable methods.

[0023] Processing continues to selectively process the region configured for the PMOS transistor **100A**, while leaving the region configured for the NMOS transistor **100B** unaffected. For example, a mask layer **350** is formed to cover the region configured for the NMOS transistor **100B** while

leaving the region configured for the PMOS transistor **100A** open. Accordingly, subsequent processing affects only the region configured for the PMOS transistor **100A**. Referring to block **20** of FIG. **2A** and FIG. **5**, portions of the fin structure **108A** adjacent to and exposed by the dummy gate structures **240** (e.g. in the source/drain regions **102**) are at least partially recessed (or etched away) by process **500A** to form the source/drain trenches **204A**. Meanwhile, the portions of the fin structure **108A** underneath the dummy gate structures **240** remain intact. Additional mask elements (not shown) may also be employed to protect areas not designed to be removed during the process **500A**. In the depicted embodiment, the process **500A** removes not only the exposed portions of fin structure **108A**, but also a portion of the underlying substrate **105**. Accordingly, the source/drain trenches **204A** extends below the top surface of the substrate **105**. In some embodiments (not shown), the process **500A** removes only the exposed portions of the fin structure **108A**, such that the top surface of the substrate **105** is exposed in the source/drain trenches **204A**. Accordingly, the source/drain trenches **204A** extends to a depth along the top surface of the substrate **105**. In some other embodiments (not shown), the process **500A** removes only some, but not all, of the fin structure **108A** adjacent the dummy gate structures **240**, such that the substrate **105** is not exposed in the source/drain trenches **204A**. Accordingly, the source/drain trenches **204A** extends to a depth above the top surface of the substrate **105**. The process **500A** may include multiple lithography and etching steps, and may use any suitable methods, such as dry etching and/or wet etching.

[0024] The formation of the source/drain trenches **204A** exposes sidewalls of the stack of semiconductor layers **110** and **120** in the region configured for the PMOS transistor **100A**. Referring to block **22** of FIG. **2B** and FIG. **6**, portions of the semiconductor layers **110** are removed through the exposed sidewall surfaces in the source/drain trenches **204A** via a selective etching process, such as process **520A**. The process **520A** is designed to remove portions of the semiconductor layers **110** but only minimally affect the semiconductor layers **120**. For example, two side portions of the semiconductor layers **110** may be removed to form openings **205** while the side portions of the semiconductor layers **120** directly above and below the openings **205** are substantially preserved. In some embodiments, those side portions of the semiconductor layers **120** may have their thicknesses reduced by about 1% to about 10% during the process **520A**. The selective etching process may be any suitable processes. In an embodiment, the semiconductor layers **120** include Si and the semiconductor layers **110** include SiGe. The process **520A** may be a wet etching process, such as a Standard Clean 1 (SC-1) solution. The SC-1 solution includes ammonia hydroxide (NH₄OH), hydrogen peroxide (H₂O₂), and water (H₂O). The SiGe semiconductor layers **110** may be etched away in the SC-1 solution at a substantially faster rate than the Si semiconductor layers **120**. The etching duration is adjusted such that the size of the removed portions of SiGe layers is controlled. As a result, desired portions of the semiconductor layers **110** are removed while the semiconductor layers **120** are only minimally affected. The optimal condition may be reached by additionally adjusting the etching temperature, dopant concentration, as well as other experimental parameters.

[0025] In another embodiment, the semiconductor layers **120** include SiGe and the semiconductor layers **110** includes Si. A cryogenic deep reactive ion etching (DRIE) process may be used to selectively etch away the Si semiconductor layer **110**. For example, the DRIE process may implement a sulfur hexafluoride-oxygen (SF₆-O₂) plasma. The optimal condition may be reached by adjusting the etching temperature, the power of the Inductively Coupled Plasma (ICP) power source and/or Radio Frequency (RF) power source, the ratio between the SF₆ concentration and the O₂ concentration, the dopant (such as boron) concentrations, as well as other experimental parameters. For example, the etching rate of a Si semiconductor layer **110** using a SF₆-O₂ plasma (with approximately 6% O₂) may exceed about 8 μm/min at a temperature of about -80° C.; while the SiGe semiconductor layers **120** are not substantially affected during the process.

[0026] The openings **205A** formed during the process **520A** extend the source/drain trenches **204A**

into areas beneath the semiconductor layers **120** and gate spacer layers **200** and/or **202**. As described above, the extent to which the semiconductor layers **110** are laterally recessed (or the size of the portion removed) is determined by the processing conditions such as the duration that the semiconductor layers **110** is exposed to an etching chemical. In the depicted embodiments, the duration is controlled such that the opening **205A** has a depth **330A** along the X-direction. In some embodiments, the etching process conditions cause the openings **205A** to have curved surfaces. For example, as illustrated in FIG. **6**, the openings **205A** may have a curved interface (surface **410**) with the remaining portions of the semiconductor layers **110**. The surface **410** protrudes from the openings **205A** into the semiconductor layers **110**. Accordingly, the openings **205A** may have a larger depth at its mid-height along the Z-direction than at its top or bottom interfaces with the semiconductor layers **120**. In such embodiments, the depth **330A** represents the average depth of the opening **205A**. The depth **330A** determines the upper limit of the lateral widths of inner spacers to be formed within the openings **205A**. In some embodiments, the depth **330A** may be about 5.5 nm to about 11 nm.

[0027] Additionally, the sidewalls of the semiconductor layers **120** may also have curved surfaces. As described above, although the semiconductor layers **120** are largely preserved during the process **520** due to their etching resistance to the etching chemical, they nevertheless may have their thicknesses slightly reduced. In addition, their sidewalls may be altered. For example, prior to the process **520A**, the semiconductor layers **120** may have substantially straight side surfaces (see FIG. **5**). But after the process **520A**, the semiconductor layers **120** may have rounded (or curved) surfaces. The surface profiles of the semiconductor layers **120** carry significances in affecting the quality of epitaxial source/drain features subsequently formed in the openings **205A**. This is described in more detail later. In the depicted embodiment, the semiconductor layers **120** each have a rounded surface **400** protruding into the source/drain trenches **204A**. In other words, the semiconductor layers **120** may have a larger length along the X-direction at its mid-height along the Z-direction than at its top or bottom interfaces with the semiconductor layers **110** (and the openings **205**). In some embodiments, the maximum length of the semiconductor layers **120** at their respective mid-height is about 20 nm to about 300 nm.

[0028] Referring to block **24** of FIG. **2B** and FIG. **7**, a dielectric material **306A** is deposited into both the source/drain trenches **204A** and the openings **205A**. The dielectric material **306A** may be selected from SiO₂, SiON, SiOC, SiOCN, or combinations thereof. In some embodiments, the proper selection of the dielectric material may be based on its dielectric constant (as described later). The deposition of the dielectric material **306A** may be any suitable methods, such as CVD, PVD, PECVD, MOCVD, ALD, PEALD, or combinations thereof. In the depicted embodiment, the dielectric material **306A** may have a top surface that extends along a top surface of the dummy gate structure **240**. In other embodiments, the dielectric material **306A** may have a top surface that extends above the top surface of the dummy gate structure **240**, and a CMP process may be performed to planarize the top surfaces of the device, and to expose the top surfaces of the dummy gate structure **240**.

[0029] Referring to FIG. **8**, the dielectric material **306A** is partially etched back by process **540A** to form new source/drain trenches **208A**. The partial etching-back completely removes the dielectric materials **306A** within the original source/drain trenches **204A**, and removes a portion, but not all, of the dielectric materials **306A** within the original openings **205A** (compare FIG. **6**). The dielectric materials **306A** remaining in the openings **205A** become the inner spacers **206A**. Accordingly, the inner spacers **206A** are formed between vertically adjacent semiconductor layers **120**. In an embodiment, the etching-back is a self-aligned anisotropic dry-etching process, such that the gate spacer layers **200** and/or **202** are used as the masking element. Alternatively, a different masking element (e.g. a photoresist) may be used.

[0030] The inner spacers **206A** interface the remaining portions of the semiconductor layers **110** at the surface **410**. Additionally, the inner spacers **206A** have a new surface **420** exposed in the

source/drain trenches **208A**. The distance between the surfaces **410** and **420** defines the lateral width **340A** of the inner spacers **206A**. The new surface **420** may be of a same or different profile as that of the surface **410**. When the surfaces **410** and **420** have different profiles, the inner spacers **206A** may have varying lateral widths at different heights of the inner spacers **206A** along the Z-direction. In such embodiments, the lateral width **340A** represents an average lateral width of the inner spacers **206A**. The lateral width **340A** is smaller than the depth **330A** of the opening **205A**. The lateral width **340A** is important to tuning the capacitance parameters; and the surface profiles are important to improving the quality of the epitaxial source/drain features subsequently formed. These aspects of the disclosure will be described in detail later.

[0031] Referring to block **26** of FIG. 2B and FIG. 9, epitaxial source/drain features **210A** are formed in the source/drain trenches **208A**. Accordingly, the epitaxial source/drain features **210A** interface with the inner spacers **206A** at the surface **420**. Multiple processes, including etching and growth processes, may be employed to grow the epitaxial source/drain features **210A**. In the depicted embodiment, the epitaxial source/drain features **210A** have top surfaces that extend higher than the top surface of the topmost semiconductor layer **120**. However, in other embodiments, the epitaxial source/drain features **210A** may alternatively have top surfaces that are about even with the top surface of the topmost semiconductor layer **120**. In some embodiments, the epitaxial source/drain features **210A** may merge together, for example, along the Y-direction (see e.g., FIG. 1A), to provide a larger lateral width than an individual epitaxial source/drain feature.

[0032] In the depicted embodiment, the epitaxial source/drain features **210A** are configured to be part of the PMOS transistor **100A**. Accordingly, the epitaxial source/drain features **210A** may include any suitable p-type semiconductor materials, such as Si, SiGe, Ge, SiGeC, or combinations thereof. The epitaxial source/drain features **210A** may be doped in-situ or ex-situ. For example, the epitaxially grown SiGe source/drain features may be doped with boron. One or more annealing processes may be performed to activate the dopants in the epitaxial source/drain features **210A**. The annealing processes may comprise rapid thermal annealing (RTA) and/or laser annealing processes. As illustrated in FIG. 9, each pair of epitaxial source/drain features **210A** are connected by multiple semiconductor layers **120**, which serve as the carrier conduit between the epitaxial source/drain features **210A** during the operation. Additionally, referring to FIG. 10, the embodiments of method **10** optionally include forming a contact etch stop layers **220A** over the epitaxial source/drain features **210A**, as well as gate spacer layers **200** and/or **202**.

[0033] The processing steps described above form epitaxial source/drain features **210A** and inner spacers **206A** of the PMOS transistor **100A**. Embodiments of method **10** continues to form components of the NMOS transistor **100B**. For example, the mask layer **350** over the region configured for the NMOS transistor **100B** is removed, and a new mask layer **360** is formed over the region configured for the PMOS transistor **100A**. Accordingly, subsequent processing affects only the region configured for the NMOS transistor **100B**. The mask layer **360** may have a material the same as or different from that of the mask layer **350**.

[0034] Referring to block **28** of FIG. 2B and FIG. 11, portions of the fin structure **108B** adjacent to and exposed by the dummy gate structures **240** (e.g. in the source/drain regions **102**) are at least partially recessed (or etched away) by process **500B** to form the source/drain trenches **204B** for subsequent epitaxial source and drain growths. The recess process **500B** may be substantially similar to the process **500A** described above with respect to FIG. 5. Accordingly, the source/drain trenches **204B** may extend to a depth that is above, along or below the top surface of the substrate **105**. The depth of the source/drain trenches **204B** may be the same as, greater than, or less than the depth of the source/drain trenches **204A**. For example, the source/drain trenches may have different depths in order to accommodate for the different growth requirements in subsequent epitaxial deposition processes in different regions, and/or to effectuate different strains in the later-formed source/drain features based on design requirements. Meanwhile, the portions of the fin structure **108B** underneath the dummy gate structures **240** remain intact. Additionally, the process **500B**

exposes the sidewalls of the stack of semiconductor layers **110** and **120**. Referring to block **30** of FIG. **2B** and FIG. **12**, portions of the semiconductor layers **110** are removed through the exposed sidewall surfaces in the source/drain trenches **204B** via a selective etching process, such as process **520B**, thereby forming openings **205B** between adjacent layers of the semiconductor layers **120**. [0035] In the depicted embodiments, the process **520B** removes a portion of semiconductor layers **110** that has a lateral width (or average lateral width) **330B**. In some embodiments, the openings **205B** may have curved surfaces. For example, as illustrated in FIG. **12**, the openings **205B** may have a curved interface (surface **440**) with the remaining portions of the semiconductor layers **110**. The surface **440** protrudes from the openings **205B** into the semiconductor layers **110**. Accordingly, the openings **205B** may have a larger depth at its mid-height along the Z-direction than at its top or bottom interfaces with the semiconductor layers **120**. In such embodiments, the depth **330B** represents the average depth of the opening **205B**. Additionally, as described above with respect to PMOS transistor **100A**, the side surfaces of the semiconductor layers **120** may be slightly etched during the process **520B** removing the semiconductor layers **110**. As a result, the semiconductor layers **120** may now have rounded (or curved surfaces). For example, as illustrated in FIG. **12**, the semiconductor layers each have a rounded surface **430** protruding into the source/drain trenches **204B**. Additionally, the portions of the semiconductor layers **120** directly above and below the openings **205B** may further have reduced thicknesses, as compared to before the process **520B** is performed. For example, these portions may have thicknesses that are about 1% to about 10% less than before the process **520B** is performed.

[0036] The process **520B** may be any suitable processes. In an embodiment, the processes **520A** and **520B** are the same method. For example, both the processes **520A** and **520B** are SC-1 etching process. Alternatively, both the processes **520A** and **520B** are cryogenic DRIE processes. In another embodiment, the processes **520A** and **520B** are different methods. For example, process **520A** may employ an SC-1 wet etching to form the openings **205A** while process **520B** may employ a cryogenic DRIE process to form the openings **205B**. In some embodiments, different etching methods result in different surface profiles, such that the profile of the surface **420** (of the openings **205A**) and the profile of the surface **440** (of the openings **205B**) are different from each other. The significances of the depths and the profiles of the surface **440** will be described in detail later.

[0037] At this stage, embodiments **10A** and **10B** of the method **10** diverge, and the processing may proceed along either of the two pathways. FIG. **2B** and FIGS. **13-21** illustrate subsequent steps of embodiment **10A** of method **10**, while FIG. **2C** and FIGS. **23-29** illustrate subsequent steps of embodiment **10B** of method **10**.

[0038] Referring to block **32A** of FIG. **2C** and FIG. **13**, a dielectric material **306B** is deposited into both the source/drain trenches **204B** and the openings **205B**. The dielectric material **306B** may be selected from SiO₂, SiON, SiOC, SiOCN, or combinations thereof. The dielectric material **306B** may be deposited by any suitable methods, such as those described above with respect to the dielectric material **306A**. The dielectric material **306B** may be the same material as, or a different material from, the dielectric material **306A**, depending on the design needs. In some embodiments, the proper selection of the dielectric material may be based on the dielectric constants. In an embodiment, this dielectric material **306B** may have a dielectric constant lower than that of the dielectric material **306A**. In some other embodiments, this dielectric material **306B** may have a dielectric constant higher than that of the dielectric material **306A**. This aspect of the dielectric material **306A** will be further discussed later.

[0039] Referring to block **34A** of FIG. **2C** and FIG. **14**, the dielectric material **306B** is etched back in process **540B** to form inner spacers **206B**. The formation of the inner spacers **206B** share many similar aspects as the formation of inner spacers **206A** (see description above with respect to FIG. **8**). For example, the process **540B** also forms new source/drain trenches **208B** (similar to the source/drain trenches **208A** described above); the inner spacers **206B** have a lateral width **340B** (similar to the lateral width **340A** of the inner spacers **206A**); the inner spacers **206B** interface with

the semiconductor layers **110** at the surface **440** (similar to the surface **410** of the inner spacers **206A**) and interface with the source/drain trenches **208B** at the surface **450** (similar to the surface **420** of the inner spacers **206A**). The process **540B** may use the same etching method as, or a different etching method from, that of the process **540A**. In the depicted embodiment, the process **540B** employs a different etching method as that of the process **540A**. For example, the process **540A** may use a wet-etching method, while the process **540B** may use a dry-etching method; or the process **540A** may use a SC-1 wet etching method, while the process **540B** may use a standard clean-2 (SC-2) wet etching method. As described above, different etching methods may produce different surface profiles. For example, the surface **450** may have a different profile from that of the surface **420**. In the depicted embodiment, the surface **450** has a smaller curvature than the surface **420**. Additionally, the lateral width **340B** may also differ from the lateral width **340A** of the inner spacers **206A**. The significances of the lateral widths and surface profiles are described in more details later.

[0040] Referring to block **34A** of FIG. **2C** and FIG. **15**, embodiment **10A** of the method **10** continues to form epitaxial source/drain features **210B** in the source/drain trenches **208B**. Accordingly, the epitaxial source/drain features **210B** interface with the inner spacers **206B** at the surface **450**. The epitaxial source/drain features **210B** generally resemble the epitaxial source/drain features **210A**, and may be formed by similar methods. However, the epitaxial source/drain features may include different dopants. In the depicted embodiment, the epitaxial source/drain features **210B** are configured for an NMOS transistor **100B**. Accordingly, the epitaxial source/drain features **210** are doped with n-type dopants in-situ or ex-situ with the epitaxial growth. For example, the epitaxially grown Si source/drain features may be doped with carbon to form silicon: carbon (Si:C) source/drain features, phosphorous to form silicon: phosphor (Si:P) source/drain features, or both carbon and phosphorous to form silicon carbon phosphor (SiCP) source/drain features. Referring to FIG. **16**, an optional contact-etch stop layer **220B** may be formed over the epitaxial source/drain features **210B**.

[0041] Referring to FIG. **17**, the mask layer **360** is removed from the region configured for the PMOS transistor **100A**, and an interlayer dielectric (ILD) layer **230** is formed over the epitaxial source/drain features **210A** and **210B**, as well as vertically over the isolation features **150**. The ILD layer **230** may also be formed in between the adjacent dummy gate structures **240** along the X-direction, and in between the adjacent source/drain features **210A/210B** along the Y-direction. The ILD layer **230** may include a dielectric material, such as a high-k material, a low-k material, or an extreme low-k material. For example, the ILD layer **230** may include SiO.sub.2, SiOC, SiON, or combinations thereof. The materials of the ILD layer **230** may be the same as or different from that of the mask layers **350** and/or **360**. The ILD layer **230** may include a single layer or multiple layers, and may be formed by a suitable technique, such as CVD, ALD, and/or spin-on techniques. After forming the ILD layer **230**, a CMP process may be performed to remove excessive ILD layer **230** and planarized the top surface of the ILD layer **230**. Among other functions, the ILD layer **230** provides electrical isolation between the various components of the PMOS transistors **100A** and NMOS transistor **100B**.

[0042] Referring to block **36A** of FIG. **2C** and FIG. **18**, the dummy gate structures **240** in the region configured for the PMOS transistor **100A** are selectively removed in process **560A**, and the dummy gate structures **240** in the region configured for the NMOS transistor **100B** are selectively removed in process **560B**. The removal of the dummy gate structures **240** creates gate trenches **242A** and **242B**, respectively, which expose the respective top surfaces and the side surfaces of the semiconductor stacks (along the Y direction). The processes **560A** and **560B** each may be selected from any suitable lithography and etching processes. In some embodiments, the lithography process may include forming a photoresist layer (resist), exposing the resist to a pattern, performing a post-exposure bake process, and developing the resist to form a masking element, which exposes a region including the dummy gate structures **240**. Then, the dummy gate structures **240** are

selectively etched through the masking element. In some other embodiments, the gate spacer layers **200** and/or **202** may be used as the masking element or a part thereof. The processes **560A** and **560B** may be the same or different processes, and may be performed at the same or different times. Accordingly, the gate trenches **242A** and **242B** may be formed at the same or different times.

[0043] Referring to blocks **38A** and **40A** of FIG. 2C and FIG. 19, the remaining portions of the semiconductor layers **110** are selectively removed through the exposed side surfaces of the semiconductor layers **120**. For example, the semiconductor layers **110** in the region configured for the PMOS transistor **100A** is removed in process **580A**; and the semiconductor layers **110** in the region configured for the NMOS transistor **100B** is removed in process **580B**. Similar to processes **520A** and **520B**, the semiconductor layers **120** may be slightly affected during the processes **580A** and **580B**. For example, the thickness of the semiconductor layers **120** may be reduced by about 1% to about 10%. The removal of the remaining portions of the semiconductor layers **110** form suspended semiconductor layers **120**, as well as openings **244A** and **244B**, respectively, in between the vertically adjacent layers. Accordingly, the center portions of the semiconductor layers **120** each have exposed top and bottom surfaces. In other words, the center portions of each of the semiconductor layers **120** are now exposed circumferentially around the X-direction. The openings **244A** may have a length **600A**; and the openings **244B** may have a length **600B**. In the depicted embodiments, the surfaces of the openings **244A** and **244B** are curved. Accordingly, the openings **244A** and **244B** each have different lengths at different heights along the Z-direction. Therefore, the lengths **600A** and **600B** represent the average lengths of the openings **244A** and **244B**, respectively. The lengths **600A** and **600B** may be the same as, or different from, each other. This is described in more details below.

[0044] The processes **580A** and **580B** may be any suitable etching processes. Further, the processes **580A** and **580B** may be the same or different processes, and may be performed at the same or different time. In addition to exposing top and bottom surfaces of the center portions of the semiconductor layers **120**, these processes also expose the sidewall surfaces of the inner spacers **206A** and **206B** respectively. The exposed sidewall surfaces may or may not have been modified by the processes **580A** and **580B**. For example, the process **580A** exposes surface **460** of the inner spacer **206A**. The surface **460** may be the same as, or different from, the surface **410**, depending on the choices of the etching methods and/or the materials of the inner spacers **206A**. Similarly, the process **580B** exposes sidewall surface **470** of the inner spacer **206B**; and the surface **470** may be the same as, or different from, the surface **440**, depending on the choices of the etching methods and/or the materials of the inner spacers **206B**.

[0045] Referring to block **42A** of FIG. 2C and FIG. 20, a gate structure **250A** is formed in the openings **244A** and the gate trenches **242A**. For example, a gate dielectric layer **246A** is formed over and between the semiconductor layers **120**, and a conductive metal layer **248A** is formed over and between the portions of the gate dielectric layers **246A**. In some embodiments, the gate dielectric layer **246A** includes multiple layers. For example, the gate dielectric layer **246A** may include a high-k dielectric layer. The high-k gate dielectric layer may be formed conformally such that it at least partially fills the gate trenches **242A**. In some embodiments, the high-k gate dielectric layer may be formed around the exposed surfaces of each of the semiconductor layers **120**, such that it wraps around each of the semiconductor layers **120** in 360 degrees. The high-k gate dielectric layer may further be formed over the side surfaces of the inner spacers **206A**, and the gate spacer layers **200**. The high-k gate dielectric layer may contain a dielectric material having a dielectric constant greater than a dielectric constant of SiO₂, which is approximately 3.9. For example, the high-k gate dielectric layer may include hafnium oxide (HfO₂), which has a dielectric constant in a range from about 18 to about 40. As various other examples, the high-k gate dielectric layer may include ZrO₂, Y₂O₃, La₂O₃, Gd₂O₃, TiO₂, Ta₂O₅, HfErO, HfLaO, HfYO, HfGdO, HfAlO, HfZrO, HfTiO, HfTaO, SrTiO, or combinations thereof. The formation of the high-k gate dielectric layers may be by any

suitable processes, such as CVD, PVD, ALD, or combinations thereof.

[0046] In some embodiments, the gate dielectric layer **246A** further includes an interfacial layer. The interfacial layer is formed between the semiconductor layers **120** and the high-k dielectric layer. Any suitable methods may be used to form the interfacial layer, such as ALD, CVD, or other deposition methods. Alternatively, the interfacial layer may also be formed by an oxidation process, such as thermal oxidation or chemical oxidation. In this instance, no interfacial layer is formed on the sidewalls of the inner spacers **206A** or the gate spacer layers **200**. In many embodiments, the interfacial layer improves the adhesion between the semiconductor substrate and the subsequently formed high-k dielectric layer. However, in some embodiments, the interfacial layer is omitted.

[0047] The conductive metal layer **248A** is formed over the gate dielectric layer **246A** and fills the remaining spaces of the gate trenches **242A** and the openings **244A**. The conductive metal layer **248A** may include a work function metal layer. In the depicted embodiment, the conductive metal layer **248A** is configured for the PMOS transistor **100A**. Accordingly, the work function metal layer may include any suitable p-type work function metal materials, such as titanium nitride (TiN), ruthenium (Ru), iridium (Ir), osmium (Os), rhodium (Rh), or combinations thereof. The conductive metal layer **248A** may further include a fill metal layer. The fill metal layer may include any suitable materials, such as aluminum (Al), tungsten (W), copper (Cu), cobalt (Co), nickel (Ni), platinum (Pt), ruthenium (Ru), or combinations thereof. In some embodiments, a CMP is performed to expose a top surface of the ILD layer **230**. The gate dielectric layer **246A** and the conductive metal layer **248A** collectively form the gate structure **250A**. The gate structure **250A** engages multiple semiconductor layers **120** to form multiple gate channels. The multiple gate channels each have a channel length (or an average channel length) that is the same as the length **600A** of the openings **244A**. Accordingly, the length **600A** is also referred to as the channel length **600A**.

[0048] Referring to block **44A** of FIG. 2C and FIG. 20, a gate structure **250B** is formed in the openings **244B** and the gate trenches **242B**. The gate structure **250B** may have similar structures as gate structure **250A**. For example, the gate structure **250B** may include a gate dielectric layer **246B** and a conductive metal layer **248B**. The gate dielectric layer **246B** may include a high-k dielectric layer (e.g., HfO₂) and an interfacial layer. The conductive metal layer **248B** may include a work function metal layer and a fill metal layer. In the depicted embodiment, the conductive metal layer **248B** is configured for the NMOS transistor **100B**. Accordingly, the work function metal layer may include any suitable n-type work function metal materials, such as titanium (Ti), aluminum (Al), tantalum (Ta), titanium aluminum (TiAl), titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbide nitride (TaCN), tantalum silicon nitride (TaSiN), or combinations thereof. The fill metal layer may include any suitable materials such as Al, W, Cu, Co, Ni, Pt, Ru, or combinations thereof. The gate structure **250B** engages multiple semiconductor layers **120** to form multiple gate channels. The multiple gate channels each have a channel length (or an average channel length) that is substantially the same as the length **600B** of the openings **244B**. Accordingly, the length **600B** is also referred to as the channel length **600B**. The length **600B** may be the same as, or different from, the length **600A**.

[0049] As describe above, the inner spacers **206A** and the inner spacers **206B** may have the same or different materials, the same or different lateral widths, and the same or different surface profiles; and the PMOS transistor **100A** may have the same or different gate channel lengths as the NMOS transistor **100B**. The materials and the lateral widths of the inner spacers may be tuned to achieve overall balanced capacitance of the nano-sheet-based CMOS; the gate channel lengths of may be tuned to achieve overall balanced current of the nano-sheet-based CMOS; and the surface profiles of the inner spacers may be tuned to optimize the qualities of the epitaxial source/drain features.

[0050] More specifically, a nano-sheet-based CMOS device includes various device regions that may be considered as capacitors. For example, the inner spacers **206A** may be considered to be the dielectric medium of a capacitor between a pair of vertically aligned conductive plates, i.e., the

sidewall of the epitaxial source/drain feature **210A** and the sidewall of the gate structure **250A**. This is denoted as region **260A** in FIG. **20**. Similarly, the inner spacers **206B** may be considered to be the dielectric medium of another capacitor between another pair of vertically aligned conductive plates, i.e., the sidewall of the epitaxial source/drain feature **210B** and the sidewall of the gate structure **250B**. This is denoted as region **260B** in FIG. **20**. In some embodiments, the optimal performance of the nano-sheet-based CMOS device may be reached when the capacitances of the regions **260A** and **260B** are balanced. Balancing the capacitances require separately tuning parameters for each of the capacitors.

[0051] In a parallel plate capacitor, the capacitance is proportional to the dielectric constant of the dielectric medium, and is inversely proportional to the separation distance of plates, according to the following equation:

$$[00001]C = \frac{\epsilon}{d} = k \epsilon_0 \frac{A}{d}$$

[0052] wherein C is the capacitance of the capacitor, ϵ is the permittivity of the dielectric medium, ϵ_0 is the permittivity of vacuum, A is the area of the plates, d is the separation distance of the plates, and k is the dielectric constant of the dielectric medium. Therefore, a smaller dielectric constant leads to a smaller capacitance; and a smaller separation distance leads to larger capacitance. The method of the present disclosure provides the flexibility of forming inner spacers of different material compositions and/or lateral widths, such that the regions **260A** and **260B** may be individually optimized for the overall optimized performances of the nano-sheet-based CMOS.

[0053] In an embodiment, the inner spacers **206A** has a lateral width **340A** of about 5 nm to about 7 nm. If the lateral width is too small, such as less than 5 nm, the protection provided to the epitaxial source/drain features **210A** in subsequent etching processes may be insufficient; if the lateral width is too large, such as greater than 7 nm, the additional cost of chip footprint may not be worth the performance enhancement. By contrast, the inner spacers **206B** has a lateral width **340B** of about 0.5 nm to about 5 nm. In other words, the inner spacer **205A** has a lateral width **340A** that may be equal to or greater than the lateral width **340B** of inner spacer **206B** by up to about 6.5 nm. If the lateral width is too small, such as less than 0.5 nm, the inner spacer dimension may suffer poor reliability; if the lateral width is too large, such as greater than 6.5 nm, it may be too close to the lateral width of the inner spacer **206A**, such that a balance of capacitance may not be reached. In an embodiment, the inner spacers **206A** includes a dielectric material that has a dielectric constant smaller than the dielectric material of the inner spacers **206B**. In some embodiments, a ratio of the lateral width **340A** to the lateral width **340B** may be about 14 to 1.1. If the ratio is too small, such as less than 1.1, the benefit afforded by the balanced capacitance may not justify the extra manufacturing cost associated with fabricating the asymmetric inner spacer structures; if the ratio is too large, such as greater than **14**, either device **100A** will have to occupy unnecessarily large chip space or the device **100B** would suffer fabrication challenges due to its small inner spacer dimension.

[0054] In some embodiments, the optimal performance of the nano-sheet-based CMOS also requires overall balanced current flow. For example, a PMOS transistor of a nano-sheet-based CMOS employs holes as the charge carriers, and an NMOS transistor of the same nano-sheet-based CMOS employs electrons as the charge carriers. Because the holes have much smaller mobility than the electrons, a shorter migration path (and therefore a shorter channel length) in the PMOS transistor than that in the NMOS transistor may be required in order to reach the same current flow level, thereby maximizing the maximum available drain current (I_{on}). In an embodiment, for example, in a memory cell, the channel length **600A** of the PMOS transistor **100A** is about 5 nm to about 15 nm. If the channel length is too small, such as less than 5 nm, the gate control of the channel region may be weakened such that leakage may increase; if the channel length is too long, such as greater than 15 nm, the additional cost of chip footprint may not be worth the performance enhancement. By contrast, the channel length **600B** of the NMOS transistor **100B** is about 10 nm to about 20 nm. In an embodiment, the channel length **600B** is greater than the channel length **600A**.

by about 5 nm to about 10 nm. If the channel length is too small, such as less than 10 nm, it may be too close to the channel length of the PMOS transistor **100A**, such that a balance of current may not be reached; if the channel length is too large, such as greater than 20 nm, the additional cost of chip footprint may not be worth the performance enhancement. In some embodiments, a ratio of the channel length **600B** to the channel length **600A** may be about 1.3 to 4. If the ratio is too small, the benefit afforded by the balanced current may not justify the extra manufacturing cost associated with fabricating the asymmetric inner spacer structures; if the ratio is too large, such as greater than 4, either device **100B** will have to occupy unnecessarily large chip space or the device **100A** would suffer fabrication challenges (such as forming gate components) due to its small channel dimension.

[0055] FIG. **21A** and FIG. **21B** are expanded cross-sectional views of the regions **260A** and **260B**, respectively, illustrating the dimensions of the surface profiles. In some embodiments, each of the semiconductor layers **120** in the region **260A** have the same surface profile, such as the profile for the surface **400**. The semiconductor layers **120** may each further have a flat portion of the outer surface, denoted as the portion **800A**. As illustrated in FIG. **21A**, the surface **400** separates from the portion **800A** at the separation point **700A**. In some embodiments, the separation point **700A** is part of a separation line that extends along the Y-direction. The surface **400** also has a tip point **710A**. The horizontal distance along the X-direction between the tip point **710A** and the separation point **700A** of the same semiconductor layer **120** is the maximum lateral width **610A** of the surface **400**. In an embodiment, the maximum lateral width **610A** is about 0.5 nm to about 4.0 nm. The vertical distance along the Z-direction between the tip point **710A** and the separation point **700A** of the same semiconductor layer **120** is about half of the thickness **620A** of the semiconductor layer **120**. In an embodiment, the thickness **620A** is about 3 nm to about 8 nm. The semiconductor layer **120** interfaces with the inner spacer **206A** at the interface point **730A**. In some embodiments, the interface point **730A** is part of an interface line that extends between the semiconductor layer **120** and the inner spacers **206A** along the Y-direction. In some embodiments, the semiconductor layer **120** also interfaces the epitaxial source/drain features **210A** at the same interface point **730A**. As illustrated, the semiconductor layers **120** each protrude into the epitaxial source/drain features **210A**. For example, referring back to FIG. **8**, the process **540A** may etch away proper amount of the dielectric material **306A**, such that the semiconductor layers **120A** each extend more towards the source/drain trenches **208A**. The horizontal distance **630A** between the tip point **710A** and the interface point **730A** represent the extent the semiconductor layers **120** protrudes into the epitaxial source/drain features as compared to the inner spacers **206A**. It is understood that a proper horizontal distance **630A** improves growth condition for the epitaxial source/drain feature **210A**. For example, epitaxial source/drain features are known to grow on semiconductor substrates but not on dielectric materials (such that the dielectric materials of the inner spacers **206A**). Accordingly, the curved portion of the semiconductor layers **120** protruding into the source/drain trenches **208A** provide a larger surface area for improved epitaxial growths. The horizontal distance **630A** may be about 0.5 nm to about 2.0 nm.

[0056] As illustrated in FIG. **21A**, the tip point **710A** and the interface point **730A** define a portion of the surface **400** denoted as **810A**. The curvature of the portion **810A** may be described by the vertical separation distance **640A** between the portion **810A** and the portion **800A** along the Z-direction at the midpoint position along the X-direction between the tip point **710A** and the interface point **730A**. In an embodiment, the vertical separation distance **640A** is about 0.2 nm to about 2.0 nm.

[0057] As described above, the inner spacer **206A** may have a curved surface **420** facing the epitaxial source/drain features **210A**. The surface **420** may have a tip point **740A**. The curvature of the surface **420** may be described by the horizontal distance **650A** along the X-direction between the interface point **730A** and the tip point **740A**, as compared to the thickness **670A** of the gate structure between the semiconductor layers **120**. In an embodiment, the horizontal distance **650A** is

less than about 2 nm. The thickness **670A** of the gate structure between the semiconductor layers **120** are largely determined by the thickness of the original semiconductor layer **110**. In other words, the thickness **670A** is similar to the thickness **300**. In an embodiment, the process **580A** slightly etches away the semiconductor layer **120**. Therefore, the thickness **670A** is slightly greater than the thickness **300**. For example, the thickness **670A** is about 1% to about 10% greater than the thickness **300**. In an embodiment, the thickness **670A** is about 8 nm to about 15 nm.

[0058] The inner spacer **206A** may also have a curved surface **460** facing the gate structure **250A**. In other words, the inner spacer **206A** may have a concave surface **460** protruding towards the gate structure **250A** along the X-direction. Accordingly, the gate structure has a concave end surface **460** that protrudes inwardly along the X-direction. The surface **460** interfaces with the portion **800A** at an interface point **720A**. In some embodiments, the interface point **720A** may be part of an interface line that extends between the semiconductor layer **120** and the inner spacer **206A** along the Y-direction. The surface **460** may also have a tip point **750A**. The curvature of the surface **460** may be described by the horizontal distance **660A** along the X-direction between the interface point **720A** and the tip point **750A**, as compared to the thickness **670A** of the gate structure **250A** between the adjacent semiconductor layers. In an embodiment, the horizontal distance **660A** is less than about 2 nm. The curvature of the surface **460** may be the same as or different from the curvature of the surface **420**. When the curvatures are the same, the inner spacers **206A** has uniform lateral width **340A** across its height; when the curvatures are different, the inner spacers **206A** has varying lateral widths across its height along the Z-direction, and the lateral width **340A** represents the average lateral width.

[0059] The portion **810A** and the surface **420** (the interface between the inner spacer **206A** and the epitaxial source/drain features **210A**) intersect at the interface point **730A** (or the interface line that extends through the interface point **730A**) and form an angle **900A**. As depicted, the angle **900A** is greater than 90°. The inventors of the present disclosure have discovered that, when the angle **900A** is smaller than or about equal to 90°, it is difficult to properly fill the entire space between the portion **810A** and the surface **420** with epitaxial materials. Accordingly, voids (or seams) may be formed in that region. By contrast, the present embodiment of the method **10** allows the angle **900A** to be greater than 90°. Therefore, the epitaxial source/drain features **210A** more easily and more reliably fill the entire space spanning the angle **900A** without leaving voids (or seams). In some embodiments, the profile of the surface **400** is at least partially determined by the process **520A** (see FIG. 6). As described above, although the semiconductor layers **120** have small etching rate during the process **520A**, the process **520A** nevertheless reshapes the profiles of the semiconductor layers **120**, resulting in the curved surface **400**. Accordingly, the profile of the surface **400** may be tuned by selecting the appropriate etching methods and adjusting etching parameters. For example, a wet etching process and/or a high-temperature chemical etching process may result in a more curved surface; while a low-temperature etching process may result in less curved surface.

[0060] With respect to the region **260B**, it shares many similar parameters and characteristics as the region **260A**. For example, the semiconductor layers **120** may have curved surfaces **430**, and the inner spacers **206B** may have curved surfaces **450** and **470**, interfacing with the epitaxial source/drain features **210B** and gate structure **250B**, respectively. The surfaces **430**, **450**, and **470** may have the same profile or different profiles between each other. For example, the surfaces **430**, **450**, and **470** may have the same curvatures or different curvatures among them. Additionally, the surface **430** may have the same curvature as, or different curvature from, that of the surface **400**; the surface **450** may have the same curvature as, or different curvature from, that of the surface **420**; the surface **470** may have the same curvature as, or different curvature from, that of the surface **460**. In the depicted embodiment, the surfaces **430** and **470** have the same curvature as the surfaces **400** and **460**, respectively. Accordingly, various aspects for these surfaces (such as denotations and dimensions) are similar between the regions **260A** and **260B**. For example, the maximum lateral width **610B** may be about the same as the maximum lateral width **610A**; the

thickness **620B** may be the same as the thickness **620A**; the semiconductor layers **120** in region **260B** also protrude into the epitaxial source/drain features **210B** as they protrude into the epitaxial source/drain features **210A**; the gate structure **250B** also has a concave end surface **470** that protrudes inwardly along the X-direction. However, in the depicted embodiment, the surface **450** has a different curvature from that of the surface **420**. This is described in more detail below.

[0061] As depicted, the inner spacer **206B** interfaces with the epitaxial source/drain feature **210B** at the surface **450**. Additionally, the inner spacer **206B** interfaces with the semiconductor layer **120** at the interface point **730B**. In some embodiments, the interface point **730B** is part of an interface line that extends between the semiconductor layer **120** and the inner spacers **206B**. In some embodiments, the semiconductor layer **120** also interfaces the epitaxial source/drain features **210B** at the same interface point **730B**. The surface **450** may have a tip point **740B**. The curvature of the surface **450** may be described by the horizontal distance **650B** along the X-direction between the interface point **730B** and the tip point **740B** of the same semiconductor layer **120**, as compared to the thickness **670B** of the gate structure **250B** between the adjacent semiconductor layers. In an embodiment, the horizontal distance **650B** is less than about 2 nm. In the depicted embodiment, the horizontal distance **650B** is smaller than the horizontal distance **650A**, while the thickness **670B** of the gate structure between the channel layers is about the same as the thickness **670A**. In other words, the curvature of the surface **450** is smaller than the curvature of the surface **420**. However, in other embodiments, the curvature of the surface **450** may be larger than or about equal to that of the surface **420**. Additionally, the curvature of the surface **450** is also smaller than the curvature of the surface **470** in the depicted embodiment. However, in other embodiments, the curvature of the surface **450** may be larger than or about equal to the curvature of the surface **470**. As described above, the distance between the surfaces **470** and **450** is the lateral width **340B** of the inner spacers **206B**. When the curvatures of the surfaces **470** and **450** are different, the lateral width **340B** represents the average lateral width of the inner spacers **206B**. In an embodiment, the lateral width **340B** is less than about 5 nm. Accordingly, the lateral width **340B** of the NMOS transistor **100B** may be smaller than the lateral width **340A** of the PMOS transistor **100A**.

[0062] Similar to corresponding features of region **260A**, the portion **810B** (defined by the tip point **710B** and the interface point **730B** on the surface **430**) and the surface **450** (the interface between the inner spacer **206B** and the epitaxial source/drain features **210B**) intersect at the interface point **730B** (or the interface line that extends through the interface point **730B**) and form an angle **900B**. As depicted, the angle **900B**, is greater than 90°. Accordingly, like the situation with respect to the angle **900A**, much of the difficulty in filling sharp angles with epitaxial semiconductor material is avoided, and no voids (or seams) are formed in the region spanning the angle **900B**. As described above, the profile of the surface **430** may be tuned during the process **520B** (see FIG. 6) by selecting the appropriate etching methods and adjusting etching parameters. For example, a wet etching process and/or a high temperature chemical etching process may result in a more curved surface, while low temperature etching process may result in a less curved surface. Accordingly, the curvature of the surface **430** as formed is smaller than the curvature of the surface **400**.

[0063] The disclosure above describes the embodiment **10A** of the method **10**. The embodiment **10B** of the method **10** shares the same processing steps **12-30** of embodiment **10A**, as illustrated in FIG. 2A. Further, FIG. 2C and FIGS. 22-27 illustrate processing steps **32B-44B**, which may be different from the embodiment **10A**. In other words, embodiment **10B** of the method **10** proceeds from the processing step **30** to **32B** and through **44B**. Because these subsequent steps **32B-44B** share many same or similar characteristics as the processing steps **32A** through **46A** already described above, the descriptions below are abbreviated for simplicity and conciseness.

[0064] Referring to block **32B** of FIG. 2C and FIG. 22, epitaxial source/drain features **210B** are formed in the source/drain trenches **204B** (illustrated in FIG. 12). There are some differences from the embodiment **10A** of the method **10**. In the embodiment **10A**, a dielectric material **306A** is deposited into the source/drain trenches **204B** before forming the epitaxial source/drain features

210A; and a portion of the dielectric material **306A** later forms the inner spacer **206A**. By contrast, these steps are omitted in the embodiment **10B**. In other words, no inner spacer is formed in embodiment **10B**. Accordingly, the epitaxial source/drain features **210B** directly interface with the semiconductor layers **110** at the surface **440**. In other aspects, the epitaxial source/drain features **210B** here generally resemble the epitaxial source/drain features **210B** of embodiment **10A** (and resemble the epitaxial source/drain features **210A** except that they may be doped with an n-type dopant rather than a p-type dopant). Additionally, the processing step of block **34B** of FIG. 2D resembles the processing step of **32A** of FIG. 2B. Also similar to the embodiment **10A**, an optional contact etch stop layer **220** may be formed over the epitaxial source/drain features **210B** (see FIG. 23).

[0065] Furthermore, the processing steps **34B-44B** of FIG. 2C of embodiment **10B** generally resemble the processing steps **36A-46A** of FIG. 2B of embodiment **10A**. For example, an ILD layer **230** is formed over both the regions configured for the PMOS transistor **100A** and the regions configured for the NMOS transistor **100B** (see block **34B** of FIG. 2D and FIG. 24); the dummy gate structures **240** are removed from both regions by processes **760A** and **760B** and forming the gate trenches **242A** and **242B** respectively (see FIG. 25); the remaining portions of the semiconductor layers **110** are removed from both regions by processes **780A** and **780B** and forming openings **244A** and **244B**, respectively (see block **36B-38B** of FIG. 2D and FIG. 26). However, because of the absence of the inner spacer in the region configured for the NMOS transistor **100B**, the openings **244A** and **244B** may have different profiles. In an embodiment, referring to FIG. 26, the process **780A** is configured such that the process removes substantially all remaining portions of the semiconductor layers **110** without substantially affecting the inner spacers **206A**. For example, the semiconductor material of the semiconductor layers **110** may be etched away at a substantially faster rate in the selected etching chemical than the dielectric material **306A** of the inner spacers **206A**. In other words, the etching selectivity largely determines the extent of etching on the PMOS side. By contrast, the semiconductor layers **110** interface directly with the epitaxial source/drain features **210B** in the region configured for the NMOS transistor **100B**. The extent of etching there may be controlled by adjusting the duration of the etching. In the depicted embodiment, the process **780B** may result in a curved surface **480**. The curved surface **480** may face an opposite direction as the curved surface **440** (compare FIG. 22). In other words, the openings **244B** may have a width greater at the midpoint of its height along the Z-direction than at the topmost and at the bottommost height along the Z-direction. Accordingly, the curved surface **480** may include the semiconductor material of the semiconductor layers **110**, the semiconductor materials of the epitaxial source/drain features **210B**, or both. In some implementations, a small portion of the semiconductor layers **110** remain after the process **780B**. In some other implementations, the entirety of the semiconductor layers **110** are removed, along with a small portion of the epitaxial source/drain features **210B**. The average length of the openings **244B** along the X-direction, denoted as **620B**, largely determines the channel length of the NMOS transistor **100B**, as described in more details below.

[0066] Referring to blocks **40B** and **42B** of FIG. 2D and FIG. 27, a gate structure **250A** is formed in the openings **244A** and the gate trenches **242A** of the PMOS transistor **100A**; a gate structure **250B** is formed in the openings **244B** and the gate trenches **242B** of the NMOS transistor **100B**. Similar to the embodiment **10A** already discussed above, the gate structure **250A** includes a gate dielectric layer **246A** and a conductive metal layer **248A**; the gate structure **250B** includes a gate dielectric layer **246B** and a conductive metal layer **248B**. In some embodiments, these layers each include materials similar to or the same as those already described above in the context of the embodiment **10A**. The gate structure **250A** has a channel length **680A**. The channel length **680A** may be similar to, or same as, the length **600A** for embodiment **10A**. The gate structure **250B** has a channel length **680B**. The channel length **680B** may be similar to, or same as, the channel length **600B** for embodiment **10A**. Additionally, as illustrated in FIG. 27, the region **270A** includes a

capacitor, similar to the region **260A** described above with respect to embodiment **10A**; however, the region **270B** does not include a similar capacitor, which is different from the region **260B** described above in the context of embodiment **10A**. For example, the gate structure **250B** (including a conductive material) directly interfaces the epitaxial source/drain features **210B** (including another conductive material). Furthermore, the gate structure **250B** has an end concave surface **480** that protrudes outwardly, as opposed to inwardly for the gate structure **250A**, along the X-direction. It is understood that one purpose of the inner spacers is to protect the epitaxial source/drain features during subsequent processes (such as etching processes to form contact holes). However, the n-type doped epitaxial source/drain features **210B** may have a different sensitivity to such subsequent processes from the p-type doped epitaxial source/drain features **210A**. For example, without the protection from the inner spacers, the p-type doped epitaxial source/drain features **210A** may be relatively easily etched (thereby damaged) in such subsequent processes; while the n-type doped epitaxial source/drain features **210B** may be more resistant to such subsequent processes under the same condition. Accordingly, in some embodiments, the inner spacers may not be necessary and can be omitted for the NMOS transistor **100B**.

[0067] FIGS. **28A** and **28B** are expanded cross-sectional views of the regions **270A** and **270B**, respectively, illustrating the dimensions of the surface profiles. The region **270A** substantially resembles the region **260A** of embodiment **10A**; while the region **270B** differs from the region **260B**. With respect to the region **270A**, as described above for the region **260A**, the semiconductor layers **120** may have curved surfaces **400**, and the inner spacers **206A** may have curved surfaces **420** and **460**, interfacing with the epitaxial source/drain features **210A** and gate structure **250A**, respectively. The surfaces **400**, **420**, and **460** may have the same profile or different profiles. For example, the surfaces **400**, **420**, and **460** may have the same curvatures or different curvatures.

[0068] With respect to the region **270B**, the structure differs substantially from that of the region **270A** or the region **260B** of the embodiment **10A**. In some embodiments, each layer of the semiconductor layers **120** share the same surface profile as the surface **430**. In addition, the semiconductor layers **120** each have a flat portion of the outer surface, denoted as the portion **800B**. As illustrated in FIG. **28B**, the surface **430** separates from the portion **800B** at the separation point **700B**. In some embodiments, the separation point is part of a separation line that extends along the Y-direction. The surface **430** also has a tip point **710B**. The horizontal distance along the X-direction between the tip point **710B** and the separation point **700B** of the same semiconductor layer **120** is the maximum lateral width **610B** of the surface **430**. In an embodiment, the maximum lateral width **610B** is about 0.5 nm to about 4.0 nm. The vertical distance along the Z-direction between the tip point **710B** and the separation point **700B** of the same semiconductor layer **120** is about half of the thickness **620B** of the semiconductor layer **120**. In an embodiment, the thickness **620B** may be the same as the thickness **620A**. For example, the thickness **620A** is about 3 nm to about 8 nm.

[0069] The gate structure **250B** interfaces the semiconductor layers **120B** at the portion **800B**. Additionally, the gate structure **250** interfaces the epitaxial source/drain features **210B** at the surface **480**. The surface **480** and the portion **800B** diverge (or separate) from each other at the separation point **720B**. In some embodiments, the separation point **720B** is part of a separation line that extend between the surface **480** and the portion **800B** along the Y-direction. Additionally, the surface **480** has a tip point **740B**. The horizontal distance between the separation point **720B** and the tip point **740B** is the maximum lateral width **660B** of the surface **480**. The curvature of the surface **480** is described by the maximum lateral width **660B** as compared to the thickness **670B** of the gate structure **250B**. As illustrated, the semiconductor layers **120** protrude into the epitaxial source/drain features **210B** such that they each extend wider than the gate structures **250B**.

[0070] As illustrated in FIG. **28B**, the tip point **710B** and the separation point **720B** define a portion of the surface **430** denoted as **810B**. The curvature of the portion **810B** may be described by the vertical separation distance **640B** between the portion **810B** and the portion **800B** along the Z-

direction at the midpoint position along the X-direction between the tip point **710B** and the separation point **700B**. In an embodiment, the vertical separation distance **640B** is about 0.2 nm to about 2.0 nm.

[0071] As described above, the gate structure **250B** may have a curved surface **480** facing the epitaxial source/drain features **210B**. The surface **480** may have a tip point **740B**. The curvature of the surface **480** may be described by the horizontal distance **660B** along the X-direction between the separation point **720B** and the tip point **740B** of the same semiconductor layer **120**, as compared to the thickness **670B** of the gate structure **250B** between the semiconductor layers **120**. In an embodiment, the horizontal distance **660B** is less than about 2 nm. The thickness **670B** of the gate structure between the semiconductor layers **120** are largely determined by the thickness of the original semiconductor layer **110**. In other words, the thickness **670B** is similar to the thickness **300** and the thickness **670A**. In an embodiment, the thickness **670B** is about 8 nm to about 15 nm.

[0072] The portion **800B** and the surface **480** intersect at the separation point **720B** and form an angle **900B**. As depicted, the angle **900B** is less than 90°. The angle **900B** is partially determined by the profile of the surface **480**, which is at least partially determined by the process **780B** (see FIG. 26). Accordingly, the magnitude of the angle **900B** may be tuned by selecting the appropriate etching methods and adjusting etching parameters. For example, a wet etching process may result in more curved surface, while a gas phase chemical etching process may result in less curved surface.

[0073] Though not intended to be limiting, embodiments of the present disclosure offer benefits for semiconductor processing and semiconductor devices. For example, the disclosed method allows the PMOS transistor and the NMOS transistor of a nano-sheet-based CMOS device to include inner spacers of different materials, dimensions, and/or surface profiles. For another example, the disclosed method allows the PMOS transistor to include an inner spacer, while the NMOS transistor to not include an inner spacer. Thus, the PMOS and NMOS transistors are afforded great flexibility in tuning the parameters to optimize its performance in order to balance with each other. As a result, better capacitance balance and/or better current balance may be achieved for the nano-sheet-based CMOS device. As such, the present disclosure provides methods that improve the performance, functionality, and/or reliability of nano-sheet-based transistors.

[0074] In an exemplary aspect, the present disclosure is directed to a semiconductor device. The semiconductor device includes a semiconductor substrate, a first pair of source and drain features and a second pair of source and drain features over the semiconductor substrate, a first stack of semiconductor layers and a second stack of semiconductor layers, a first gate and a second gate. The first pair of source and drain features are p-type doped; and the second pair of source and drain features are n-type doped. The first stack of semiconductor layers connect the first pair of source and drain features along a first direction. The second stack of semiconductor layers connect the second pair of source and drain features along a second direction. The first gate has a first portion between vertically adjacent layers of the first stack of semiconductor layers. The second gate has a second portion between vertically adjacent layers of the second stack of semiconductor layers. The first portion has a first dimension along the first direction, and the second portion has a second dimension along the second direction. The second dimension is larger than the first dimension.

[0075] In some embodiments, the first portion has a concave end surface protruding inwardly along the first direction, and the second portion has a concave end surface protruding outwardly along the second direction. In some embodiments, the semiconductor device further includes a first inner spacer. The first inner spacer has a third dimension along the first direction and connects the first portion and one of the first pair of source and drain features. In some embodiments, a layer of the first stack of semiconductor layers includes a region where a material of the layer, a material of the first inner spacer, and a material of the one of the first pair of source and drain features are interfaced with one another. In some embodiments, the fourth dimension is smaller than the third dimension. In some embodiments, the first dimension is about 5 nm to about 15 nm, the second

dimension is about 10 nm to about 20 nm, the third dimension is about 5 nm to about 7 nm, and the fourth dimension is about 0.5 nm to about 5 nm. In some embodiments, the first inner spacer includes a first material, the second inner spacer includes a second material, and the second material is different from the first material. In some embodiments, the first inner spacer has a first interface with the one of the first pair of source and drain features; the second inner spacer has a second interface with the one of the second pair of source and drain features. The first interface has a first surface profile, the second interface has a second surface profile. The second surface profile is different from the first surface profile. In some embodiments, the first stack of semiconductor layers are stacked along a third direction perpendicular to the first direction. A first layer of the first stack of semiconductor layers has a rounded end portion and a middle portion. The rounded end portion protrudes from an edge of the first inner spacer into the one of the first pair of source and drain features along the first direction by a first length. The first length is about 0.5 nm to about 2 nm. The middle portion is in between and directly contacts two adjacent portions of the first gate. The rounded end portion has a first thickness along the third direction at the midpoint of the first length. The middle portion has a second thickness along the third direction. A first thickness difference between the first thickness and the second thickness is about 1 nm to about 4 nm. In some embodiments, the second stack of semiconductor layers are stacked along a fourth direction perpendicular to the second direction. A second layer of the second stack of semiconductor layers has a rounded end portion and a middle portion. The rounded end portion protrudes from an edge of the second inner spacer into the one of the second pair of source and drain features along the second direction by a second length. The middle portion is in between and directly contacting two adjacent portions of the second gate. The rounded end portion has a third thickness along the fourth direction at the midpoint of the second length. The middle portion has a fourth thickness along the fourth direction. A second thickness difference between the third thickness and the fourth thickness is different from the first thickness difference.

[0076] In an exemplary aspect, the present disclosure is directed to a method. A structure is received. The structure includes a semiconductor substrate, a first stack of first semiconductor layers and second semiconductor layers over the semiconductor substrate, a second stack of third semiconductor layers and fourth semiconductor layers over the semiconductor substrate, a first dummy gate structure over the first stack, and a second dummy gate structure over the second stack. The first semiconductor layers and the second semiconductor layers have different material compositions and alternate with one another within the first stack. The third semiconductor layers and the fourth semiconductor layers have different material compositions and alternate with one another within the second stack. A first portion of the first stack on two sides of the first dummy gate is removed to form a first pair of source/drain trenches, thereby exposing a pair of first side surface of the first stack. A first portion of the first semiconductor layers is removed from the exposed pair of first side surfaces to form first gaps. A first inner spacer is formed in the first gaps, where the first inner spacer has a first dimension along a direction connecting the first pair of source/drain trenches. A first pair of source/drain features are epitaxially grown in the first pair of source/drain trenches. A first portion of the second stack on two sides of the second dummy gate is removed to form a second pair of source/drain trenches, thereby exposing a pair of second side surfaces of the second stack. A first portion of the third semiconductor layers is removed from the exposed second side surfaces to form second gaps. A second inner spacer is formed in the second gaps, where the second inner spacer has a second dimension along a direction connecting the second pair of source/drain trenches. The second dimension is different from the first dimension. A second pair of source/drain features are epitaxially grown in the second pair of source/drain trenches. The first and the second dummy gate structures are removed to form a first gate trench over the first stack and a second gate trench over the second stack, respectively. The second portion of the first semiconductor layers are removed from the first gate trench. A second portion of the third semiconductor layers is removed from the second gate trench. A first gate electrode is formed

in the first gate trench. A second gate electrode is formed in the second gate trench.

[0077] In some embodiments, the epitaxially growing of the first pair of source/drain features includes epitaxially growing the first pair of source/drain features and doping the first pair of source/drain features with an n-type dopant; the epitaxially growing of the second pair of source/drain features includes epitaxially growing the second pair of source/drain features and doping the second pair of source/drain features with a p-type dopant; and the forming of the second inner spacer includes forming the second inner spacer with the second dimension larger than the first dimension. In some embodiments, the removing of the first portion of the first semiconductor layers includes removing with a first etching method; the removing the first portion of the third semiconductor layers includes removing with a second etching method. The second etching method differs from the first etching method. In some embodiments, the forming of the second inner spacer includes forming the second inner spacer with a material different from a material of the first inner spacer. In some embodiments, the forming of the second inner spacer includes forming the second inner spacer with a surface profile different from a surface profile of the first inner spacer. In some embodiments, the epitaxially growing of the first pair of source/drain features includes growing on a first interface between the first inner spacer and a layer of the first semiconductor layers such that one of the first pair of source/drain features directly contacts the first interface; and the epitaxially growing of the second pair of source/drain features includes growing on a second interface between the second inner spacer and a layer of the third semiconductor layers such that one of the second pair of source/drain features directly contacts the second interface. In some embodiments, the forming of the first gate electrode includes forming the first gate electrode in contact with surfaces of the first inner spacer, and the forming of the second gate electrode includes forming the second gate electrode in contact with surfaces of the second inner spacer.

[0078] In an exemplary aspect, the present disclosure is directed to a method. A structure is received. The structure includes a semiconductor substrate, a first stack of first semiconductor layers and second semiconductor layers over the semiconductor substrate, a second stack of third semiconductor layers and fourth semiconductor layers over the semiconductor substrate, a first dummy gate structure over the first stack, and a second dummy gate structure over the second stack. The first semiconductor layers and the second semiconductor layers have different material compositions and alternate with one another within the first stack. The third semiconductor layers and the fourth semiconductor layers have different material compositions and alternate with one another within the second stack. A first portion of the first stack on two sides of the first dummy gate structure is removed to form a first pair of source/drain trenches, thereby exposing a pair of first side surfaces of the first stack. A first portion of the first semiconductor layers is removed from the exposed pair of first side surfaces to form first gaps. First inner spacers are formed in the first gaps. The first inner spacers each has a first dimension along a first direction connecting the first pair of source/drain trenches. A first pair of source/drain features are epitaxially grown in the first pair of source/drain trenches, such that the first pair of source/drain features each directly contacts with one of the first inner spacers. A first portion of the second stack on two sides of the second dummy gate structure is removed to form a second pair of source/drain trenches, thereby exposing a pair of second side surfaces of the second stack. A first portion of the third semiconductor layers is removed from the exposed pair of second side surfaces to form second gaps. A second pair of source/drain features are grown in the second pair of source/drain trenches, such that the second pair of source/drain features each directly contacts with a side surface of a second portion of the third semiconductor layers. The first and the second dummy gate structures are removed to form a first gate trench over the first stack and a second gate trench over the second stack, respectively. A second portion of the first semiconductor layers is removed from the first gate trench. The second portion of the third semiconductor layers is removed from the second gate trench. A first gate electrode is formed in the first gate trench. A second gate electrode is formed in the second gate trench.

[0079] In some embodiments, the forming of the first gate electrode includes forming a first concaved gate electrode end surface that protrudes inwardly along the first direction; and the forming of the second gate electrode includes forming a second concaved gate electrode end surface that protrudes outwardly along the first direction. In some embodiments, the forming of the first gate electrode includes forming the first gate electrode in contact with surfaces of the first inner spacers, and the forming of the second gate electrode includes forming the second gate electrode in contact with surfaces of the second pair of source/drain features.

[0080] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device, comprising: a first stack of semiconductor layers over a semiconductor substrate; a second stack of semiconductor layers over the semiconductor substrate; a first source/drain feature connected to the first stack of semiconductor layers; a second source/drain feature connected to the second stack of semiconductor layers; a first gate structure engaging with the first stack of semiconductor layers and having a first curved sidewall surface profile; and a second gate structure engaging with the second stack of semiconductor layers and having a second curved sidewall surface profile, wherein the first source/drain feature includes a p-type dopant, the second source/drain feature includes an n-type dopant, the first curved sidewall surface profile has a first curvature, and the second curved sidewall surface profile has a second curvature, and the first curvature is greater than the second curvature.
2. The device of claim 1, wherein the first gate structure spans a first width along a first direction, the second gate structure spans a second width along the first direction, and the second width is greater than the first width.
3. The device of claim 1, further comprising: first inner spacers laterally between the first gate structure and the first source/drain feature along a first direction; and second inner spacers laterally between the second gate structure and the second source/drain feature along the first direction, wherein the first inner spacers span a greater width along the first direction than that of the second inner spacers.
4. The device of claim 3, wherein the first inner spacers have a first sidewall surface and a second sidewall surface each having the first curved sidewall surface profile.
5. The device of claim 3, wherein the second inner spacers have a first sidewall surface having the second curved sidewall surface profile and a second sidewall surface having a third curved sidewall surface profile different from the second curved sidewall surface profile.
6. The device of claim 5, wherein the third curved sidewall surface profile has a third curvature less than the second curvature.
7. The device of claim 1, further comprising: first inner spacers laterally between the first gate structure and the first source/drain feature along a first direction; and second inner spacers laterally between the second gate structure and the second source/drain feature along the first direction, wherein the first inner spacers include a first dielectric material, the second inner spacers include a second dielectric material, and the first dielectric material has a smaller dielectric constant than the second dielectric material.
8. The device of claim 1, further comprising: first inner spacers laterally between the first gate

structure and the first source/drain feature along a first direction, the first inner spacers vertically sandwiched between semiconductor layers of the first stack of semiconductor layers, wherein the first stack of semiconductor layers protrudes into the first source/drain feature along the first direction past outer surfaces of the first inner spacers, wherein an angle between where the first inner spacers interface the protruding portions of the first stack of semiconductor layers is greater than 90 degrees.

9. The device of claim 8, wherein the first inner spacers have first curved surfaces that curve inwards toward the first gate structure, wherein the first stack of semiconductor layers have second curved surfaces that curves outward towards the first source/drain feature.

10. The device of claim 1, wherein the first gate structure includes a top portion above a topmost semiconductor layer of the first stack of semiconductor layers and a bottom portion below the topmost semiconductor layer of the first stack of semiconductor layers, wherein the top portion spans a greater length along a first direction than the bottom portion.

11. The device of claim 1, wherein the second gate structure includes a top portion above a topmost semiconductor layer of the second stack of semiconductor layers and a bottom portion below the topmost semiconductor layer of the second stack of semiconductor layers, wherein the top portion spans a smaller length along a first direction than the bottom portion.

12. A device, comprising: a first stack of semiconductor layers over a semiconductor substrate; a second stack of semiconductor layers over the semiconductor substrate; a p-type source/drain feature connected to the first stack of semiconductor layers; an n-type source/drain feature connected to the second stack of semiconductor layers; a first gate structure engaging with the first stack of semiconductor layers; a second gate structure engaging with the second stack of semiconductor layers; and inner spacers adjacent to the p-type source/drain feature without being adjacent to the n-type source/drain feature, wherein the first gate structure is laterally separated from the p-type source/drain feature by the inner spacers, and the second gate structure is in contact with the n-type source/drain feature, wherein portions of the n-type source/drain feature extends vertically between the semiconductor layers of the second stack.

13. The device of claim 12, wherein the first gate structure have outer surfaces that curve inwards toward a center of the first gate structure, wherein the second gate structure have outer surfaces that curve outwards away from the center of the first gate structure.

14. The device of claim 12, wherein the first gate structure includes a top portion above a topmost semiconductor layer of the first stack of semiconductor layers and a bottom portion below the topmost semiconductor layer of the first stack of semiconductor layers, and the top portion of the first gate structure spans a greater length along a first direction than the bottom portion of the first gate structure, wherein the second gate structure includes a top portion above a topmost semiconductor layer of the second stack of semiconductor layers and a bottom portion below the topmost semiconductor layer of the second stack of semiconductor layers, and the top portion of the second gate structure spans a smaller length along the first direction than the bottom portion of the second gate structure.

15. The device of claim 12, wherein the first gate structure spans a first width along a first direction, the second gate structure spans a second width along the first direction, and the second width is greater than the first width.

16. The device of claim 12, wherein the inner spacers have a lateral width ranging between about 5 nm to about 7 nm.

17. A device, comprising: a first stack of semiconductor layers over a semiconductor substrate; a second stack of semiconductor layers over the semiconductor substrate; a p-type source/drain feature connected to the first stack of semiconductor layers; an n-type source/drain feature connected to the second stack of semiconductor layers; a first gate structure having a top portion over the first stack of semiconductor layers and a bottom portion wrapping around each semiconductor layer of the first stack of semiconductor layers; a second gate structure having a top

portion over the second stack of semiconductor layers and a bottom portion wrapping around each semiconductor layer of the second stack of semiconductor layers; first inner spacers laterally between the bottom portion of the first gate structure and the p-type source/drain feature along a first direction; and second inner spacers laterally between the bottom portion of the second gate structure and the n-type source/drain feature along the first direction, wherein the first inner spacers have curved side surfaces of a same curvature, and the second inner spacers have curved side surfaces of a different curvature.

18. The device of claim 17, wherein the first inner spacers have a first width along the first direction, the second inner spacers have a second width along the first direction, and the first width is greater than the second width.

19. The device of claim 17, wherein the first inner spacers include a first dielectric material, the second inner spacers include a second dielectric material, and the first dielectric material has a smaller dielectric constant than the second dielectric material.

20. The device of claim 17, wherein the top portions of the first and the second gate structures span a substantially same width along the first direction, wherein the bottom portion of the first gate structure span a smaller width than a width of the second gate structure along the first direction.
